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Osculations of spectral lines in a layered medium

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HIGHLIGHTS

- Deals with the presence or absence of double roots, and near double roots, in the frequency–wavenumber spectra for a horizontally layered medium.
- Presents (in closed-form) a previously unknown set of formulas for the double roots of an elastic stratum.
- Demonstrates that no double roots can ever exist in the case of generalized SH waves.
- Posits the strong conjecture that double roots for generalized SVP waves can solely exist when layered plates and full spaces possess a geometric and material symmetry plane.
- A layered elastic half-space will never exhibit double roots.

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The second author (PM) wishes to dedicate this paper to the memory of his dear friend Peter Winkler (1934–2014), formerly of Carl Zeiss at Jena, Germany. Peter was a gifted mathematician, an avid mountain climber and a skilled photographer, and above all, a true legend among his friends. In short, he was a very special man who lived a very special life.

Keywords:
Waves in layered media
Normal modes of wave propagation
Double roots in wave spectra
Avoided crossings
Curve veering

ABSTRACT

When the dense web of spectral curves for the normal modes of wave propagation in a layered medium are examined carefully, it is often observed that pairs of curves approach each other to the point that they appear to intersect, yet a closer examination reveals that they do not. In the technical literature these are referred to as avoided crossings, curve veering, mode-splitting, double cusps, diabolical points, or *osculation points*. We consider herein their presence – or absence – in a homogeneous elastic layer underlain by an elastic half-space, and include the two extreme cases of a homogeneous stratum where the half-space is infinitely rigid, and a free plate when it is infinitely flexible. First we show rigorously that the spectral curves for generalized Love waves (SH waves) in a layered medium will not ever intersect, even if in some cases they may come close and osculate. We then demonstrate that the spectral lines for generalized Rayleigh waves (SV-P waves) in a homogeneous stratum will only intersect when Poisson's ratio is either exactly 1/4, or is in some other fractional (but not continuous) ratio below 1/3, in which case we provide the intersection points in closed form. We also provide the general conditions that must be satisfied by the SVP spectral lines for double roots to exist in a layered medium, whether they are plates, strata, half-spaces of full spaces. We then posit that intersections of SVP spectral lines (i.e. double roots) can only occur when the layered medium exhibits material symmetry with respect to some horizontal plane. Although at first sight the stratum case might seem to contradict this principle we show that ultimately it does not. Thus, neither a non-symmetric free plate nor a half-space will ever have intersecting lines. By contrast, a symmetric *full* space does have such intersections, as do also symmetrically layered plates, among which the homogeneous free plate and the homogeneous doubly-clamped plate exhibit an abundance of double roots.

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1. Introduction

Nearly a century has passed since the first publications on eigenvalue problems with "peculiar properties" made their first appearance, probably started by the two seminal and widely-quoted papers by the polymath John von Neumann and the Nobel Prize winning physicist Eugene Paul Wigner (1929) [1,2]. In their paper they considered certain types of eigenvalue problems that depended on some parameter and observed that in the vicinity of a critical value, the branches for two eigenvalues seemed to approach each other exquisitely until they nearly coalesced to form a double root – a degenerate eigenvalue problem – but then diverged abruptly as if a force of repulsion existed between the two. They referred this as an "avoided crossing". Furthermore, at the near encounter the modal shape of one mode was transferred onto the other mode as if the branches had indeed crossed and each mode simply continued normally across the junction.

In the English literature the avoided crossings are often referred to as *osculating* curves, which derives from the ancient Roman word *osculum* for "kiss with a small mouth" or "kiss on the cheek". On the other hand, the alternative and somewhat facetious designation "diabolo" is for the shape of a double cone (or double cusp) that is formed by the spectral branches as they approach their vertex. The *Encyclopedia Britannica* [3] describes the *diabolo* as a Chinese toy introduced by 18th century missionaries into Europe and known in England as *the devil in two sticks*, or as *le diable* in early 19th century France. A pellet from an air gun is also sometimes referred to as diabolo. In physics, the double cone may well be traced back to the works of Hamilton (1833) [4] who expounded on an interesting physical effect associated with coincident eigenvalues referred to as *conical refraction.* In modern times, Berry & Wilkinson (1984) [5], among others, refer to the double-cone as *diabolo* and to the degeneracies themselves as *diabolical points*.

Unbeknownst of the Neumann–Wigner paper, half a decade later the eminent seismologists Sezawa and Kanai (1935) [6] reported discovering a second Rayleigh mode for a uniform layer over an elastic half-space, both with equal mass density and Poisson's ratio 0.25, and for various stiffness ratios μ_1/μ_2 of the layer and the half-space. As is well known, for some given material configuration this is a transcendental eigenvalue problem in the frequency, with the horizontal wavenumber being the independent parameter, or vice-versa. They observed that the two spectral branches approached each other to the point of nearly touching but without crossing, and that in the vicinity of their close encounter the group velocities changed rapidly in value. In addition, they observed that the modal characteristic of one mode was somehow passed onto the other without the branches actually intersecting. Finally, they also observed that when the half-space became infinitely rigid, the curves finally touched and actually crossed. Yet in the light of what we shall prove in the ensuing, their observation was nothing but serendipitous, because the stratum has no cross-over points *except* when Poisson's ratio is either exactly equal to 0.25, or is less than $1/3$ and can be written as some specific rational fraction, as will be seen.

In the decades following the Neumann–Wigner paper the concept of *avoided crossings* seems to have resurfaced from time to time in the literature on quantum physics, solid state physics, acoustics, geophysics and mechanics, not to mention mathematics. In seismology, perhaps one of the earliest explicit references to osculation problems may be found in a book by Levshin [7]. Then again, judging by the recent increase in the number of papers on the topic of avoided crossings, it would seem that the interest of mathematicians & physicists [8–12], engineering scientists [13–16] and seismologists [17–20] may have been reawakened.

Among recent works on osculations is an interesting essay in the Internet by seismologist T. Forbriger (2006) [17], who cogitates at length about the topic of avoided crossings. In the first pages of his article he carries out a thorough review of previous works on osculating curves in seismology, to which we refer the reader for further information. Thereafter, he discusses the mathematical concepts, attempts to shed light with an example of a weakly coupled pendulum, and affirms that osculations in wave propagation problems are not ubiquitous, but actually relatively rare, and that they result from a constellation of parameters satisfying miscellaneous constraints. He reasons that osculations preferentially arise when layers of high and low stiffness alternate and act as waveguides for one and another mode, and that the impedance contrast of the intervening layers somehow insulates one mode from the other, but not perfectly, and so the modes remain weakly coupled. He concludes his interesting essay with the comment that "a physically correct solution of the coupled eigenvalue problem cannot be reduced into a simple form (i.e. equation) that would allow for a smooth transition from an uncoupled system to a weakly coupled system". He then adds "*osculation points are a rather rare occurrence*", and "*the degeneracy of the eigenvalue problem in media with depth-dependent elastic properties does not appear to be possible*", concluding with the comment "*the case studied by Sezawa and Kanai appears to constitute an exceptional example*".

In Chapter 4 of his dissertation, Tran (2009) [21] also makes allusion to osculation points, but he makes no formal distinction between avoided and actual crossings. Instead, he is mainly concerned with the horizontal to vertical (H/V) ratios of motion at the surface of various layered structures. Still, he goes on to develop expressions for double roots in a homogeneous layer with fixed bottom which are based on a dispersion equation formulated via ray theory. Remarkably, he arrives in this way at expressions (his Eqs. 4.11, 4.15, 4.16) which can be shown to agree with two of our own, rigorously derived expressions given later on for the double roots of a stratum, namely our Eqs. (7) and (8b).

It should be added that Forbriger (2006) [17], in a footnote on the first page of his essay also reported detecting numerically that a stratum with Poisson's ratio 0.25 exhibits crossing points, but that a tiny deviation of just 0.00005 from that ideal value was sufficient to suppress the crossing points. However, he did not demonstrate this analytically or prove that for some other Poisson's ratios the spectral curves might also cross. He also suggests that no formal proof yet exists concerning either the feasibility or the impossibility of double roots in a realistically layered medium.

But at least for a uniform plate, we already know that spectral lines do indeed cross. Indeed, Mindlin (1958) [22,23] fully unraveled the plate spectrum and provided *exact* equations for the infinitely many intersection points. Still, double roots are indeed rare: as we shall see later on, there are none if the layered system fails to possess a horizontal symmetry plane, of which the layered half-space is an example. But non-symmetric plates lack them too. However, while such systems may be totally devoid of any double roots, they often exhibit myriad osculating points that approach each other exquisitely to the point of masquerading as true crossing points. Indeed, osculation points are indeed abundant, perhaps if not like grains of sand in a beach, then perhaps as flowers in a blooming garden that are neatly arranged into patterns of colors, shapes and sizes.

As we shall demonstrate in this article, an abundance of osculation points with regular distribution patterns do exist even in a perfectly homogeneous stratum having just two wave guides: the free surface and the rigid base, with both being separated merely by uniform material. We contend that osculation points are both abundant and do not require complicated layering to become apparent, excepting perhaps the case of generalized Love waves, where the standard one-layer over an elastic half-space model possesses neither double roots nor osculation points, and so more layers are necessary to cause the latter to become manifest. We also shall prove rigorously that the spectrum for a homogeneous stratum has crossing points only when Poisson's ratio is less than $1/3$ and then only when ν can be written in some specific rational form. More generally, we shall also lay down the mathematical conditions that must be satisfied for double roots to exist in the case of an arbitrarily layered medium, and posit that double roots will exist solely when the layered medium possesses a horizontal plane of symmetry (both geometric and material).

We begin our analysis by reviewing briefly the concept of layer impedance, which we use to formulate the transcendental eigenvalue problems for SH and SV-P wave propagations.

2. Dynamic impedance

Consider an arbitrarily layered system subjected to plane waves contained in the x–z vertical plane. As shown by Kausel (2006) [24], when each layer is considered as a free body in space, its dynamic equilibrium for plane waves cast in the frequency–horizontal wavenumber domain can be fully characterized by a complex-valued, symmetric dynamic impedance matrix. For anti-plane (SH) motion, the external traction–displacement relationship for a single layer is of the form

$$\begin{Bmatrix} \tau_{y1} \\ \tau_{y2} \end{Bmatrix} = \begin{bmatrix} k_{11} & k_{12} \\ k_{12} & k_{11} \end{bmatrix} \begin{Bmatrix} u_{y1} \\ u_{y2} \end{Bmatrix} \tag{1}$$

where the indices 1, 2 refer to the upper and lower interfaces; τ_{y1}, τ_{y2} are the tangential *external* tractions applied in horizontal direction y at these two elevations, and u_{y1}, u_{y2} are the displacements observed at those two locations elicited by propagating or evanescent SH waves, and the k_{ij} are the elements of the SH impedance matrix given in Appendix A. An implied factor $e^{\mathrm{i}(\omega t - kx)}$ on either side of the equation has been omitted for reasons of convenience, with ω being the frequency and k the horizontal wavenumber.

On the other hand, for plane-strain SV-P waves, the dynamic equilibrium equation for a single layer is of the form

$$\begin{Bmatrix} \tau_{x1} \\ -\mathrm{i}\,\sigma_{z1} \\ \tau_{x2} \\ -\mathrm{i}\,\sigma_{z2} \end{Bmatrix} = \left[\begin{array}{cc:cc} K_{11} & K_{12} & K_{13} & K_{14} \\ K_{12} & K_{22} & -K_{14} & K_{24} \\ \hdashline K_{13} & -K_{14} & K_{11} & -K_{12} \\ K_{14} & K_{24} & -K_{12} & K_{22} \end{array}\right] \begin{Bmatrix} u_{x1} \\ -\mathrm{i}\,u_{z1} \\ u_{x2} \\ -\mathrm{i}\,u_{z2} \end{Bmatrix} \tag{2}$$

where again the sub-indices 1, 2 refer to the upper and lower bounding surfaces of the layer; τ_{xj}, σ_{zj}, $j = 1, 2$ are the tangential (horizontal) and normal (vertical) *external* tractions that must be applied to maintain equilibrium; u_{xj}, u_{zj} are the displacements observed at those interfaces, elicited by both propagating or evanescent SVP waves in the layer; the K_{ij} are the elements of the SVP impedance matrix given in Appendix A; and $\mathrm{i} = \sqrt{-1}$ is a convenience factor that renders the stiffness matrix symmetric.

The matrices in curly braces in Eqs. (1), (2) involving the elements k_{ij}, K_{ij} are the *exact* impedance matrices for the free layer in question. Its elements are given in Appendix A (Kausel, 2006) [23], which list also the impedance matrix for a homogeneous half-space that must, of course, be assembled with the material properties of the half-space. A layered system can then be modeled by appropriately overlapping the stiffness matrices of individual layers, as is done in the finite element method. The reader is cautioned, however, that the matrices herein are *exact* and thus do not introduce any discretization errors.

3. Generalized Love waves: their spectra never cross

As is well known, the propagation of horizontally polarized (SH) waves in a vertically inhomogeneous medium is characterized by equations that have the mathematical form of a Sturm–Liouville problem (SLP). These problems have been studied intensively over the last century, and mathematicians have established that the eigenvalues of such problems are never degenerate, i.e. that no double roots exist. However, when the wave guide is open as is the case of a half-space or a full space, then one or both of the boundary conditions and the associated radiation conditions are applied at an infinite depth

and the problem becomes "singular", in which case one can no longer guarantee the absence of double roots, or indeed even if roots exist at all (Malischewsky, 1987) [25]. An example of the latter is the homogeneous stratum which, no matter how deep has infinitely many "locked" SH modes. By contrast, the homogeneous half-space has no such modes. Thus, we provide herein a proof of the absence of double roots by other means.

With reference to Eq. (1) and the detailed expressions for the impedance elements $k_{ij}^{(n)}$ given in Appendix A, the transcendental eigenvalue problem for generalized Loves waves in a layered soil consisting of $N-1$ layers underlain by an elastic half-space is of the tri-diagonal and symmetric form

$$\begin{Bmatrix} k_{11}^{(1)} & k_{12}^{(1)} & 0 & 0 \\ k_{12}^{(1)} & k_{11}^{(1)}+k_{11}^{(2)} & \ddots & 0 \\ 0 & \ddots & \ddots & k_{12}^{(N-1)} \\ 0 & 0 & k_{12}^{(N-1)} & k_{11}^{(N-1)}+k_H \end{Bmatrix} \begin{Bmatrix} u_{y1} \\ u_{y2} \\ \vdots \\ u_{yN} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{Bmatrix} \tag{3}$$

where the superscript in parenthesis identifies the layer, and the stiffness elements must be set up with the properties appropriate for each layer. The spectral lines for generalized Love waves are then obtained by setting the determinant of the dynamic system matrix to zero, i.e. $\Delta = 0$. Now, for the purposes in our ensuing analysis, we proceed first to carry out a static condensation (i.e. elimination through Gaussian reduction) of all degrees of freedom below the second, after which we obtain

$$\begin{Bmatrix} k_{11}^{(1)} & k_{12}^{(1)} \\ k_{12}^{(1)} & k_{11}^{(1)}+Z \end{Bmatrix} \begin{Bmatrix} u_{y1} \\ u_{y2} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \tag{4}$$

where $Z = Z(\omega, k)$ is the impedance of the complete layered system that exists underneath the second interface. We observe in passing that if we were to set the determinant of the condensed Eq. (4) to zero, we would recover the full spectrum for the complete medium, but that is not our immediate goal. Instead, we now assume that there exists at least one pair of intersecting spectral lines, which is the same as saying that there are two coincident eigenvalues, and that requires the system matrix to have a loss of rank of two. The implication is that the matrix in Eq. (4) must be a null matrix, so all elements of that matrix must be zero, i.e. $k_{11}^{(1)} = 0$, $k_{12}^{(1)} = 0$, $k_{11}^{(1)}+Z = 0$. The first and last of these three conditions are clearly possible, but the second one is definitely impossible, because that would demand the satisfaction of $k_{12}^{(1)} = 1/\sinh ks_1h_1 = 0$ (see Appendix A), which cannot be achieved with any real or complex values of the argument (except at $k = \infty$ or $h_1 = \infty$, both of which are uninteresting).

We should add that an alternative, equivalent condition for double roots to exist is the simultaneous satisfaction of $\Delta = 0$ and $\Delta' = 0$, where the prime indicates differentiation with respect to the horizontal wavenumber k. This follows from Lancaster's Trace Theorem, $\Delta' = \Delta\,\mathrm{tr}\left(\mathbf{K}^{-1}\mathbf{K}'\right) = \mathrm{tr}\left(\mathbf{C}\,\mathbf{K}'\right)$ where $\mathbf{C}$ is Cramer's matrix of cofactors, and $\mathbf{K}' = \frac{d}{dk}\mathbf{K}$. When double roots exist, then all minor determinants of order $N-1$ also vanish so $\mathbf{C} = \mathbf{O}$ and therefore $\Delta' = 0$.

We shall now demonstrate that no intersection points will ever exist in the spectral lines for a symmetrically layered plate or for a full, symmetrically layered space subjected to generalized Love waves. There are two simple proofs.

First, the symmetric system is characterized by an eigenvalue of the form

$$\begin{Bmatrix} k_{11}^{(1)}+Z & k_{12}^{(1)} \\ k_{12}^{(1)} & k_{11}^{(1)}+Z \end{Bmatrix} \begin{Bmatrix} u_{y1} \\ u_{y2} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

in which the upper and lower half-spaces have the same impedance Z. Again, a double root would demand $k_{12}^{(1)} = 0$, which cannot be satisfied for the same reasons given before for a half-space.

A second proof relies on the recognition that a symmetric structure possesses both symmetric and anti-symmetric modes. Thus, one can split the full space (or plate) along the symmetry plane, and in the resulting lower half-space (or plate) obtain the symmetric modes by imposing a traction-free upper surface, while the anti-symmetric modes follow from imposing a constrained surface. These will result in spectral lines $\omega_j(k)$ and $\varpi_j(k)$, respectively. But the latter differs from the former by the addition of a single constraint, in which case Rayleigh's *interlacing theorem* applies (Rayleigh, 1894) [26]. This theorem states that the frequencies of the free dynamic system will interlace with those of the constrained system, i.e. $\omega_1 \le \varpi_1 \le \omega_2 \le \varpi_2 \le \omega_3 \cdots$. It follows immediately that the branches $\omega_j(k)$, $\varpi_j(k)$ cannot intersect. We conclude that the branches of generalized Love waves in a layered medium will *never* intersect, no matter what the layer properties may be. Still, one can construct layered systems where the branches come really close, i.e. they osculate, even if only at some isolated points.

4. SV-P waves in a homogeneous, doubly clamped plate

Consider a homogeneous layer of total thickness $H = 2h$ which is clamped (i.e. fixed) at both its upper and lower bounding surfaces. We refer to this entity as the double-thickness, fully clamped plate, which is materially symmetric with respect

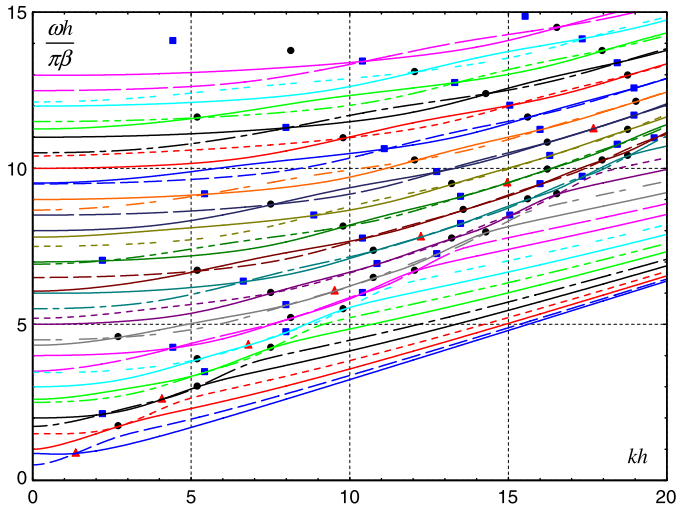


Fig. 1. First 40 spectral lines for a doubly-clamped, double-thickness plate, $\nu = 0.25$. Dashed lines = symmetric modes; solid lines = anti-symmetric modes. Markers indicate double poles: Triangles = set 1 (Eq. (5a)); circles = set 2 (Eq. (5b)); squares = set 3 (Eq. (5c)).

to its middle horizon. Being symmetric, it possesses a set of symmetric modes for which the vertical displacements and tangential stresses vanish at the mid-horizon, and a set of anti-symmetric modes where it is the horizontal displacements and the vertical stresses that vanish at that horizon. Thus, these modes can alternatively be obtained by considering a homogeneous layer of thickness h that is first bounded at its upper surface by horizontal rollers, and thereafter by vertical rollers, and in both of these cases by a rigid lower surface. With reference to Eq. (2), this is the same as saying that the layer has two complementary mixed boundary conditions (MBC) at its upper surface and is clamped at the base, the first of which is characterized by the equation $K_{11}u_{x1} = 0$ and the second by the equation $K_{22}u_{z1} = 0$, so double roots exist wherever $K_{11} = K_{22} = 0$ are satisfied simultaneously.

When these two complementary eigenvalue problems are solved in full (see Appendix B for details) and the intersections of the spectral curves are sought, one finds that the doubly clamped plate exhibit a plethora of double roots for any Poisson's ratio which are given by the following three sets of expressions:

$$\frac{\omega h}{\beta} = j\pi\,\frac{\alpha}{\sqrt{\alpha^2+\beta^2}}, \qquad kh = j\pi\,\frac{\alpha\beta}{\alpha^2+\beta^2} \quad j = 1, 2, 3\ldots \tag{5a}$$

$$\frac{\omega h}{\beta} = \pi\sqrt{2\,(1-\nu)}\sqrt{\left(n-\tfrac{1}{2}\right)^2-\left(m-\tfrac{1}{2}\right)^2}$$
$$kh = \pi\sqrt{2\,(1-\nu)}\sqrt{\left(\frac{\beta}{\alpha}\right)^2\left(n-\tfrac{1}{2}\right)^2-\left(m-\tfrac{1}{2}\right)^2} \qquad \begin{matrix} m, n = 1, 2, 3, \ldots \\ \beta\,(2n-1) > \alpha\,(2m-1) \end{matrix} \tag{5b}$$

$$\frac{\omega h}{\beta} = \pi\sqrt{2\,(1-\nu)}\sqrt{n^2-m^2}, \qquad kh = \pi\sqrt{2\,(1-\nu)}\sqrt{\left(\frac{\beta}{\alpha}\right)^2 n^2-m^2}, \qquad \begin{matrix} m, n = 1, 2, 3\ldots \\ \beta n > \alpha m \end{matrix}. \tag{5c}$$

Fig. 1 shows the frequency–wavenumber spectrum for a doubly clamped plate of thickness $2h$ and Poisson's ratio $\nu = 1/4$. The shape of the markers distinguishes between the solutions to (5a) (triangles), (5b) (circles), or (5c) (squares). Clearly, this problem exhibits an abundance of true crossings, and no osculation points. All double roots lie above the straight line with phase velocity $c = \alpha$. In particular, the double roots in Eq. (5a) (i.e. triangles) lie on the line with constant phase velocity $c = \sqrt{\alpha^2+\beta^2}$.

5. SV-P waves in a plate with mixed boundary conditions (MBC)

A plate with mixed boundary conditions is one where both the upper and lower surfaces are free to move tangentially and are constrained normally, or the other way around, i.e. they are roller boundaries. Such a plate is characterized by very simple normal modes, namely pure P waves or pure S waves that reverberate up and down while propagating laterally. This is because when either P or S waves hit a roller boundary, they reflect completely and without mode conversion. Thus, their dispersion spectra are very easy to derive and are known in closed form. Moreover, the intersections of the spectral lines of the plate with MBC represent pure P and S waves with the same horizontal wavenumber and a relative amplitude such that either the normal or the tangential displacement vanishes on both surfaces. These intersection points coincide perfectly

with Eq. (5c) for a fully clamped plate of double thickness, so they constitute a subset of the intersections for that plate (this is because of (5a) and (5b)). Alternatively, if we write instead the equations for the doubly clamped plate in terms of its true thickness $H = 2h$ (obtained by multiplying Eqs. (5a)–(5c) by two, and replacing on the left hand side $2h$ by H), then the double crossings of the doubly-clamped plate are seen to agree with the *even–even* or the *odd–odd* crossings of the plate with MBC, although the latter still misses the crossing implied by Eq. (5a).

6. SV-P waves in a homogeneous free plate

The homogeneous, free plate was thoroughly studied by Mindlin (1958) [22,23], who fully unraveled its dispersion spectrum. He demonstrated that the spectral lines of the free plate exhibit double roots which coincide with a subset of the crossing points of a plate with MBC at both surfaces, namely Eq. (5c) when m, n are both *odd* or are both *even* (but not when they are *even–odd*). Here the modes at the intersection points are again pure P and S waves with relative weights such that the reactions at the rollers vanish. Thus, they are again a subset of the intersections of the spectral curves for the plate with MBC.

Mindlin also demonstrated that when Poisson's ratio is below the threshold $\nu = 0.2631$, additional crossing points for the free plate can be found which are given by

$$k_{nj}h = \frac{2j\pi}{\sqrt{\left(\frac{C_{Rn}}{\alpha}\right)^2 - 1} + \sqrt{\left(\frac{C_{Rm}}{\beta}\right)^2 - 1}}, \qquad \frac{\omega_{nj}h}{\beta} = \frac{2j\pi\,\frac{C_{Rn}}{\beta}}{\sqrt{\left(\frac{C_{Rn}}{\alpha}\right)^2 - 1} + \sqrt{\left(\frac{C_{Rn}}{\beta}\right)^2 - 1}} \quad \begin{matrix} n = 1, 2 \\ j = 0, 1, 2, \ldots \end{matrix} \tag{6}$$

where $C_{R1} > \alpha, C_{R2} > \alpha$ are the two *false* Rayleigh roots of the well-known bi-cubic Rayleigh function, which are real when $0 \le \nu < 0.2631$ and complex otherwise. However, Mindlin himself did not present examples demonstrating these additional roots, and their expression also contained an error which was corrected by Freedman (1990) [27].

Inasmuch as each of the spectral curves of the plate are bounded by the spectral curves for the plate with MBC and that there are only two branches within each bound, it would seem that, in general, the spectral curves of the homogeneous plate never osculate, but only cross each other. An exception might be the case when Poisson's ratio slightly exceeds the critical value $\nu = 0.2631$, in which case the false Rayleigh root crossings (6) vanish and turn into osculation points. In summary, the homogeneous plate has myriad crossing points and except as noted above, no osculation points.

7. SV-P waves in a homogeneous stratum

Consider next a homogeneous stratum i.e. a homogeneous elastic layer of finite depth with a stress-free surface and underlain by a rigid base where the displacements vanish. This system is characterized by a single 2×2 impedance matrix, which after imposing the rigid boundary condition at the bottom $u_2 = 0, w_2 = 0$ as well as the vanishing of tractions at the surface, $\tau_1 = 0, \sigma_1 = 0$ leads us to a transcendental eigenvalue problem for free waves given and evaluated in detail in Appendix C. As shown there, the double roots of a stratum must simultaneously satisfy the three conditions $K_{11} = K_{22} = K_{12} = 0$. Thus, they constitute a subset of the double roots of the fully clamped plate of double thickness (which already satisfy $K_{11} = K_{22} = 0$), namely the subset (if any) that satisfies simultaneously the additional condition $K_{12} = 0$, i.e. the vanishing of the coupling term between horizontal and vertical degrees of freedom at the surface.

From Appendix C, we find that a stratum exhibits double roots in the following special circumstances:

(a) Case 1: If $\nu = 0.25$, then an infinite number of double roots exist of the form

$$kh = (2j-1)\,\pi\,\frac{\sqrt{3}}{4}, \qquad \frac{\omega h}{\beta} = (2j-1)\,\pi\,\frac{\sqrt{3}}{2}, \quad j = 1, 2, 3, \ldots, \qquad \nu = \tfrac{1}{4} \tag{7}$$

whose constant phase velocity is $c = \omega/k = 2\beta$. Of this infinite set, the fundamental mode for $j = 1$ is the most important for practical applications.

(b) Case 2: If (m, n) is a pair of even–even or odd–odd integers satisfying the condition $n > 2m - \frac{1}{2}$ and in addition the pair $2m - 1,\ 2n - 1$ is relatively prime, then there exists an infinite set of double roots of the form

$$\nu = \frac{2m-1}{2\,(n+m-1)} \quad 0 < \nu < \tfrac{1}{3}, \qquad j = 1, 2, 3 \ldots \tag{8a}$$

$$\begin{aligned} kh &= \tfrac{1}{2}\pi\,(2j-1)\,\sqrt{(2n-1)\,(2n-4m+1)} \\ &= \tfrac{1}{2}\pi\,(2j-1)\,\frac{\sqrt{(1-\nu)\,(1-3\nu)}}{\nu}, \\ \frac{\omega h}{\beta} &= \pi\,(2j-1)\,\sqrt{(2n-1)\,(n-m)} \\ &= \tfrac{1}{2}\pi\,(2j-1)\,\frac{\sqrt{2\,(1-\nu)\,(1-2\nu)}}{\nu} \end{aligned} \tag{8b}$$

Table 1
Examples of Poisson's ratios satisfying (8a) ($m, n < 10$).

m	n	ν as fraction	ν
1	9	1/18	0.055556
1	7	1/14	0.071429
1	5	1/10	0.100000
1	3	1/6	0.166667
2	6	3/14	0.214286
2	4	3/10	0.300000

Table 2
Examples of Poisson's ratios satisfying (9a) ($m, n < 10$).

m	n	ν as fraction	ν
4	7	1/99	0.010101
1	2	1/9	0.111111
2	5	13/63	0.206349
2	7	37/135	0.274074
1	4	13/45	0.288889
1	8	61/189	0.322751

whose phase velocity is

$$c = \frac{\omega}{k} = 2\beta\sqrt{\frac{n-m}{2n-4m+1}} = 2\beta\sqrt{\frac{1-2\nu}{2\,(1-3\nu)}} \tag{8c}$$

with ν given by (8a).

(c) Case 3: If (m, n) is a pair of even–odd or odd–even integers satisfying the condition $n > m\sqrt{3}$ and in addition they are relatively prime, then there exists an infinite set of double roots of the form:

$$\nu = \frac{n^2-3m^2}{3n^2-3m^2}, \quad 0 < \nu < \tfrac{1}{3}, \qquad j = 1, 2, \ldots \tag{9a}$$

$$kh = (2j-1)\,\pi\, n\frac{\sqrt{3}}{3}, \qquad \frac{\omega h}{\beta} = (2j-1)\,\pi\, n\frac{2\sqrt{3}}{3} \tag{9b}$$

whose constant phase velocity is $c = \omega/k = 2\beta$.

These three sets of solutions are mutually exclusive, that is, there are no repetitions of Poisson's ratios, even if by appropriate choice of m, n one can get really close to any Poisson's ratio less than 1/3. With few exceptions, however, the vast majority of the double roots implied by (8b), (9b) take place at very high wavenumbers and frequencies. Tables 1 and 2 show a short, albeit incomplete, list of values satisfying these equations when $m < 10$, $n < 10$, sorted by Poisson's ratio.

Remarkably, two of the three sets of double points lie along the straight line through the origin with a phase velocity $v_{ph} = 2\beta$ which does not depend on Poisson's ratio, even though the spectral lines themselves do depend on that ratio. Thus, we shall refer to this special line as the *osculatrix*. Also, it would seem at first that by choosing $n = 3m$ in Eq. (9b) one could reduce case 3 to case 1, but that is not quite the case because that combination would make the pair m, n to be both even or both odd, in contradiction to the assumptions made when developing case 3 in Appendix C.

The first set (Eq. (7)) agrees with the double roots detected numerically by Sezawa & Kanai [6], and using ray methods, by Tran [26] (who refers to the intersections as osculations), although he does not indicate that it is only valid when $\nu = 1/4$. Tran also arrived at two additional formulas for the intersection points which in combination can be shown to agree with our formulas (8b), but he completely missed our case 3 (Eq. (9b)).

Having found the preceding equations for the double roots of a homogeneous stratum in closed form, we are reasonably confident that the above set is complete and that no further double roots exist. It is rather remarkable that the infinite (but countable) set of discrete double roots given by (8b), (9b) cover densely (even if discontinuously) the interval of Poisson's ratios $0 \le \nu \le 1/3$. Left out are Poisson's ratios whose values are either irrational or transcendental numbers. Conversely, given any arbitrary but fixed value of Poisson's ratio ν, one can always find pairs of values (m, n) in formulas (8a), (9a) yielding a Poisson's ratio that is arbitrarily close to the given one, provided that $\nu < 1/3$. However, in most cases those pairs will imply spectral intersections at very high modal numbers, and thus are physically even if not mathematically suspect.

When the spectral curves for some specific Poisson's ratio are plotted, it will be found that strong osculations exist at locations which agree with a sub-set of the double roots for the double-thickness, doubly clamped plate evaluated at that same Poisson's ratio, namely Eq. (5b) evaluated for even–even or odd–odd indices m, n, or Eq. (5c) evaluated at even–odd indices. If the Poisson's ratio happens to satisfy also either of Eqs. (8a) or (9a) (or it is exactly equal to 0.25) then some of these points will be true intersections. Observe that all of these points already satisfy at least two of the three conditions required for a double root to exist in a stratum, namely $K_{11} = K_{22} = 0$.

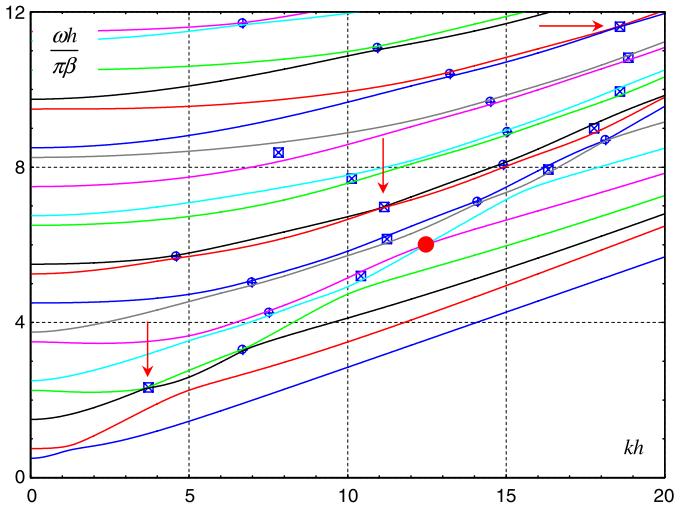


Fig. 2. Spectrum for $\nu = 1/10$. True crossing indicated by the solid dot, which agrees with Eq. (8a) when $(m, n) = 1, 5$. Open circles are from Eq. (5b) (with m, n even–even or odd–odd), while open squares are from (5c) (even–odd or odd–even). The open squares pointed at by the three arrows are sharp osculations which lie on the osculatrix and virtually agree with the double crossings for $\nu = 1/9$, which would be obtained from (9b) using $m, n = (1, 2), (3, 6), (5, 10)$.

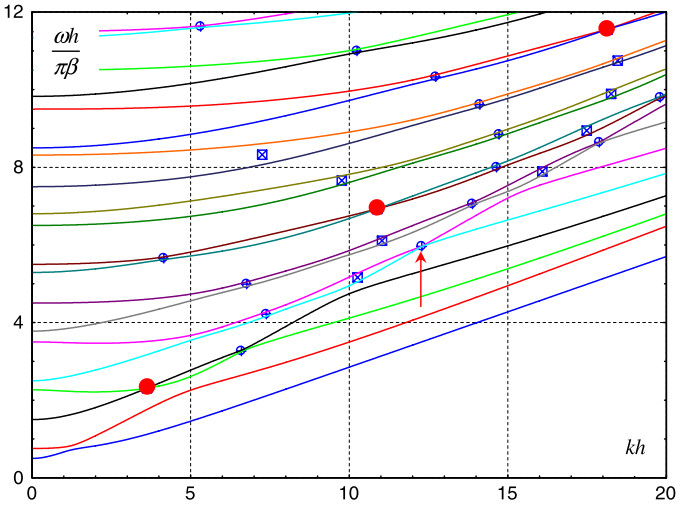


Fig. 3. Spectrum for $\nu = 1/9$. True crossings indicated by the solid dots, which agree perfectly with Eq. (9a) when $m, n = (1, 2), (3, 6), (5, 10)$ and lie on the osculatrix. Open circles are from Eq. (5b) (with m, n even–even or odd–odd), while open squares are from (5c) (even–odd or odd–even). The open circle pointed at by the arrow is a sharp osculation which virtually agrees with the double crossings for $\nu = 1/10$, which would be obtained from Eq. (8b) using $m, n = 1, 5$ (see Fig. 2).

Still, the location of both the double roots and of the osculation points is exquisitely sensitive to the value of ν, even if the spectra hardly change when this parameter is modified slightly. This is illustrated in Figs. 2 and 3, which show the spectral lines for $\nu = 1/10$ and $\nu = 1/9$. If these two sets of spectral lines were to be overlapped, one would observe that they virtually coincide. However, the double roots from the first disagree strongly from those of the second, and indeed, a double root in one figure becomes an osculation point in the other. The implication is that double roots observed for some Poisson's ratio quickly "dissolve" and turn into sharp osculation points when Poisson's ratio is slightly altered.

Having found a set of double roots for a stratum, we now go on to interpret physically these double roots, and for this purpose we consider again the related problem of a double-thickness, doubly clamped plate. As we have seen, the modes of that system agree with those of a layer with mixed boundary conditions at the upper surface, namely horizontal rollers followed by vertical rollers. Such a system is characterized by the two system of equations

$$\begin{Bmatrix} 0 \\ -\mathrm{i}\,\sigma_{z1} \end{Bmatrix} = \begin{bmatrix} K_{11} & K_{12} \\ K_{12} & K_{22} \end{bmatrix} \begin{Bmatrix} u_{x1} \\ 0 \end{Bmatrix} \quad \text{horizontal rollers} \Rightarrow K_{11} = 0 \tag{10a}$$

Table 3
Layered free plate.

Layer	Mass density	Thickness	Shear modulus	Poisson's ratio
1	1.00	0.25	1.00	0.10
2	1.00	0.50	1.50	0.00
3	1.00	0.25	1.10	0.15

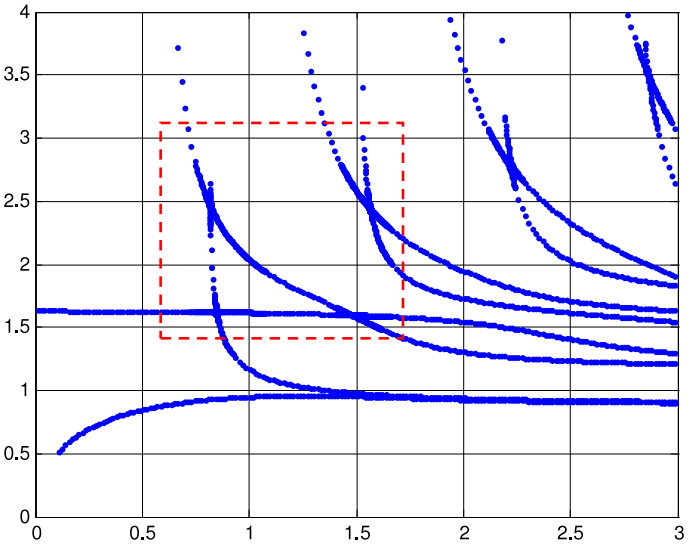


Fig. 4. Phase velocity vs. frequency (Hz), 3-layer non-symmetric plate. Figs. 5 through 7 show progressive levels of magnification, as indicated by the dashed rectangles. It is seen that the apparent intersections are actually very sharp osculations. There are no double roots.

$$\begin{Bmatrix} \tau_{x1} \\ 0 \end{Bmatrix} = \begin{bmatrix} K_{11} & K_{12} \\ K_{12} & K_{22} \end{bmatrix} \begin{Bmatrix} 0 \\ -\mathrm{i}\, u_{z1} \end{Bmatrix} \quad \text{vertical rollers} \Rightarrow K_{22} = 0 \tag{10b}$$

where $\tau_{x1} = -\mathrm{i}\, K_{12} u_{z1}$, $\sigma_{z1} = \mathrm{i}\, K_{21} u_{x1}$ are the "support reactions" at the rollers. On the other hand, for the stratum the system equation is

$$\begin{Bmatrix} 0 \\ 0 \end{Bmatrix} = \begin{bmatrix} K_{11} & K_{12} \\ K_{12} & K_{22} \end{bmatrix} \begin{Bmatrix} u_{x1} \\ -\mathrm{i}\, u_{z1} \end{Bmatrix}$$

for which the double roots require $K_{11} = K_{22} = K_{12} = 0$. Clearly, these double roots are then a subset of the roots of the doubly-clamped plate, namely those that satisfy the additional condition $K_{12} = 0$, as we have already seen. But if the coupling terms in Eqs. (10a,b) vanish, then the reactions at the rollers vanish as well, i.e. $\tau_{x1} = 0$, $\sigma_{z1} = 0$. This means that the rollers can be taken away, at least as far as the subset of intersection points satisfying the conditions $\tau_{1x} = 0$, $\sigma_{1z} = 0$ is concerned, in which case what is left is the original stratum with a free surface, and satisfying the third condition for a double root, namely $K_{12} = 0$. Hence, the double roots of the stratum coincide with a subset of the double roots of the symmetrically clamped double-thickness layer, namely those which elicit no stresses in the mid-plane, and this explains how a non-symmetric stratum could still have double roots. This fortuitous coincidence is not likely to be repeated for any other non-symmetric, layered structure because the terms in K_{11}, K_{22}, K_{12} will not have a simple relationship with each other as they do for a homogeneous plate (which allowed obtaining the closed form solution in the first place).

8. Layered plates, half-spaces and full-spaces

As a first example, consider a three-layer plate with the properties given in Table 3.

Fig. 4 shows the phase velocities for this non-symmetric case, while Figs. 5–7 depict various levels of magnification. At the macroscopic level (Fig. 4) it seems rather "obvious" that the dispersion curves intersect at various points and even repeated zooming into these areas appears to corroborate that interpretation, but after truly massive magnification (Fig. 7), we see that *all* curves veer apart at the very "last minute" and so the double roots are not such but actually are osculation points.

By way of contrast, consider next a rather similar plate, but one that is symmetric with respect to the mid-horizon, as defined in Table 4. Fig. 8 shows the phase velocities for this case, while Figs. 9, 10 zoom into the details.

Comparison with those of the previous example shows that the dispersion spectra are very similar to each other, except that now most – but not all – of the osculation points became true intersections.

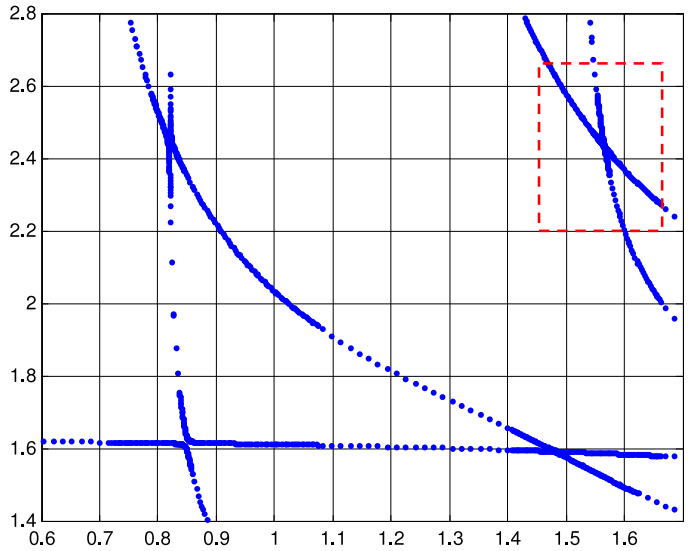


Fig. 5. First level of magnification of Fig. 4.

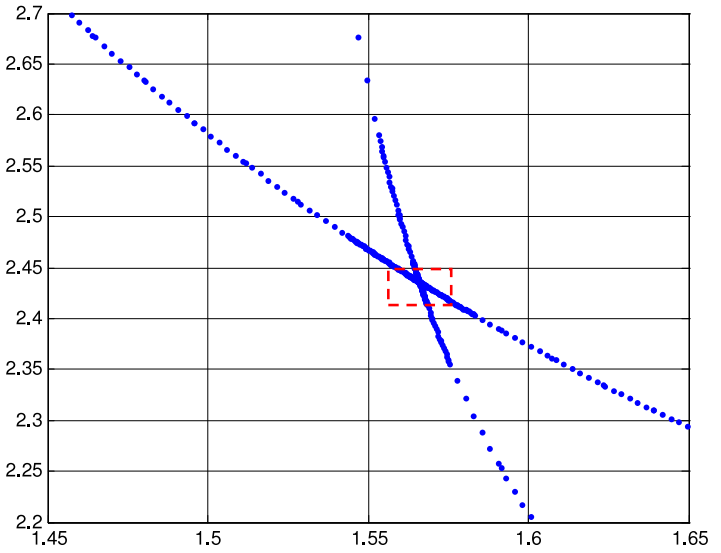


Fig. 6. Second level of magnification of Fig. 4. It still looks like a crossing.

Table 4
Symmetrically layered free plate.

Layer	Mass density	Thickness	Shear modulus	Poisson's ratio
1	1.00	0.25	1.00	0.10
2	1.00	0.50	1.50	0.00
3	1.00	0.25	1.00	0.10

Out of many other examples involving non-symmetrically layered system that we have analyzed (but which are not part of this article), including layered half-spaces, there was *not one case* with double roots, although like the non-symmetric plate, we have seen plenty of osculation points masquerading almost perfectly as intersections. It behooves next to show what happens when the symmetric system in question is not a plate or stratum of finite depth, but is a full, layered space. The properties for this case are given in Table 5.

This system is symmetric with respect to the mid-elevation of layer 3, so it possesses both symmetric and anti-symmetric modes. The dispersion spectra for these two can just as well be obtained by halving the system and considering in turn two

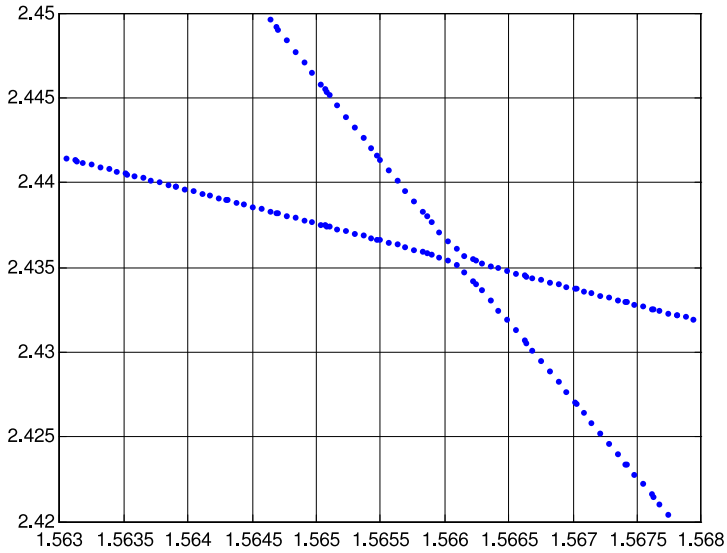


Fig. 7. Third level of magnification of Fig. 4. Aha! It is an osculation!.

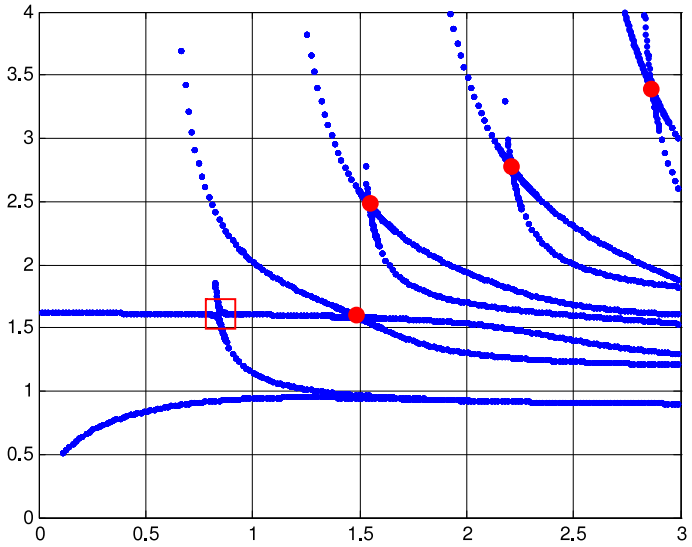


Fig. 8. Phase velocity vs. frequency (Hz), 3-layer symmetric plate. The solid dots are true intersections, the open square is an osculation point. Details for two of these are shown in Figs. 9, 10.

Table 5
Symmetrically layered full-space.

Layer	Mass density	Thickness	Shear wave velocity	Poisson's ratio
1	1.00	∞	1.50	1/3
2	1.00	1.00	1.00	1/3
3	1.00	1.00	0.50	1/3
4	1.00	1.00	1.00	1/3
5	1.00	∞	1.50	1/3

layered half-spaces with mixed boundary conditions at the upper surface, the first one FC (horizontal rollers) and the other CF (vertical rollers). Figs. 11, 12 depict the results of this operation, showing both sets of phase velocity spectra in two different colors (note: color is visible in the online version only). Clearly, the symmetric full space does exhibit intersection points, and it may also possess osculation points, although no clear one appears within the frequency interval evaluated.

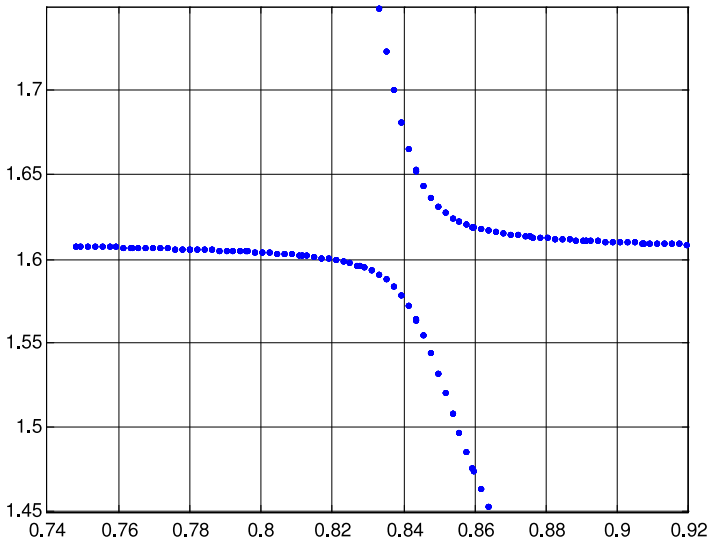


Fig. 9. Details of osculation point in Fig. 8.

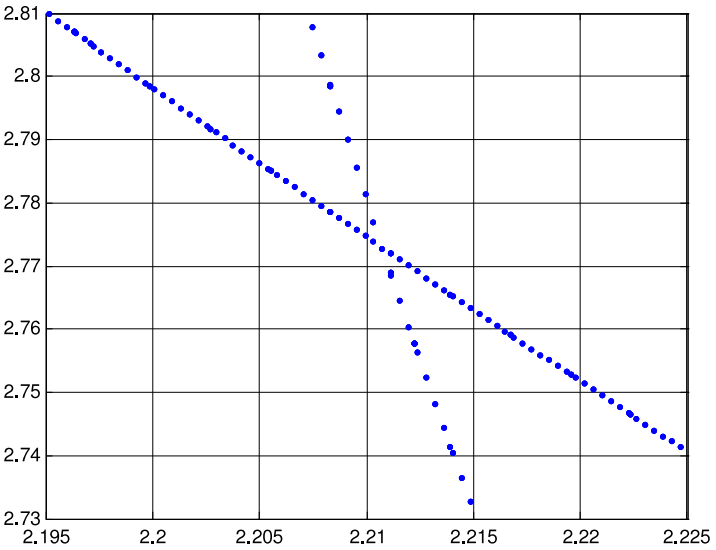


Fig. 10. Details of intersection in Fig. 8.

9. SV-P waves in layered medium: conditions for, and conjecture on double roots

The characteristic equation for a layer over an elastic half-space is of the form

$$\left\{\begin{array}{cc:cc} K_{11}^{(1)} & K_{12}^{(1)} & K_{13}^{(1)} & K_{14}^{(1)} \\ K_{12}^{(1)} & K_{22}^{(1)} & -K_{14}^{(1)} & K_{24}^{(1)} \\ \hdashline K_{13}^{(1)} & -K_{14}^{(1)} & K_{11}^{(1)}+K_{11}^{H} & -K_{12}^{(1)}+K_{12}^{H} \\ K_{14}^{(1)} & K_{24}^{(1)} & -K_{12}^{(1)}+K_{12}^{H} & K_{22}^{(1)}+K_{22}^{H} \end{array}\right\} \left\{\begin{array}{c} u_1 \\ -\mathrm{i}\, w_1 \\ u_2 \\ -\mathrm{i}\, w_2 \end{array}\right\} = \left\{\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array}\right\}$$

where the elements $K_{ij}^{(1)}$ are those for the top (i.e. first) layer, and the K_{ij}^{H} are the impedance elements of the half-space, both of which are given in Appendix A. In the case of multiple layers, the equation is block-tridiagonal as above, but with more sub-matrices and displacement components before those of the half-space. Still, for our present purposes, it suffices to consider this simpler case.

The roots of this secular equation are obtained by setting the determinant of the symmetric, global impedance matrix to zero. Thus, when k is taken as a parameter, one is lead to numerous spectral branches. More to the point, regarding the issue of intersections, these constitute double roots and thus represent a loss of rank of 2 in the characteristic equation. This means that not only must the 4×4 determinant of this equation vanish to satisfy a true root, but so must also any of the

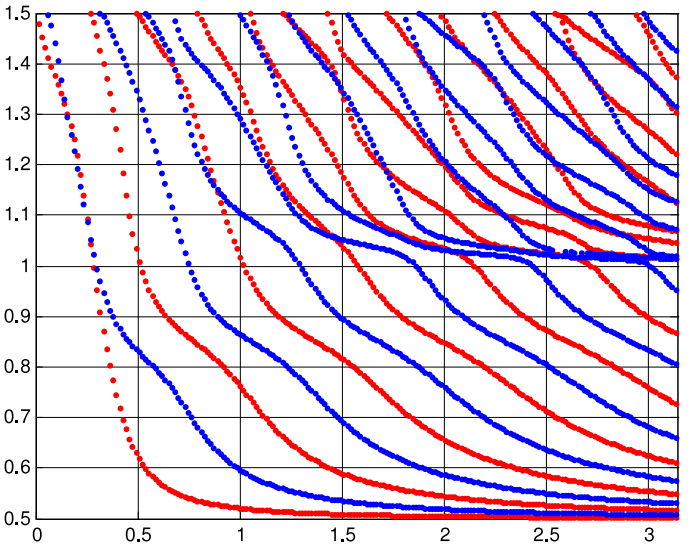


Fig. 11. Phase velocity vs. frequency (Hz) for a 5-layer full space. Blue = symmetric modes, red = antisymmetric modes. There are many intersections and no clear osculations. Details are shown in Fig. 12. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

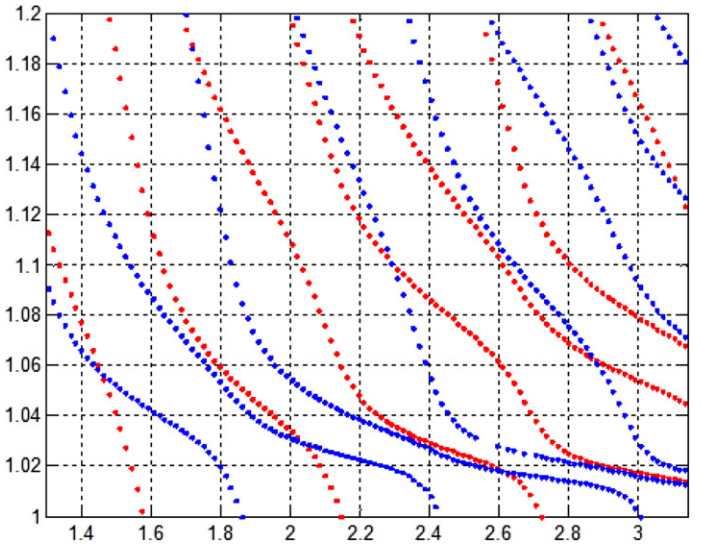


Fig. 12. Details of Fig. 11, showing the intersections of the symmetric and anti-symmetric branches. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3×3 sub-determinants that can be formed with any three rows and columns. Alternatively, if the matrix is subdivided into four 2×2 sub-matrices of the form

$$\begin{Bmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} \\ \mathbf{K}_{21} & \mathbf{K}_{22} \end{Bmatrix} \begin{Bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{Bmatrix} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{0} \end{Bmatrix}$$

and the surface degrees of freedom $\mathbf{u}_1$ are subsequently condensed out, then

$$\mathbf{Z}\mathbf{u}_2 = \left(\mathbf{K}_{22} - \mathbf{K}_{21}\mathbf{K}_{11}^{-1}\mathbf{K}_{12}\right)\mathbf{u}_2 = \mathbf{0} \tag{11}$$

where $\mathbf{Z}$ is the total impedance as perceived from the layer-half-space interface. However, because of the loss of two in rank, this matrix must also be a null matrix, i.e. $\mathbf{Z} = \mathbf{O}$—if only at the double root. Carrying out this condensation we are led to the three conditions $Z_{11} = 0, Z_{22} = 0, Z_{12} = 0$. With the shorthand $H = |\mathbf{K}_{11}| = K_{11}^{(1)}K_{22}^{(1)} - K_{12}^{(1)}K_{12}^{(1)} \neq 0$, these conditions are found to be

$$Z_{11} \to H\left(K_{11}^{H} + K_{11}^{(1)}\right) - K_{14}^{(1)}\left(K_{11}^{(1)}K_{14}^{(1)} + K_{12}^{(1)}K_{13}^{(1)}\right) - K_{13}^{(1)}\left(K_{22}^{(1)}K_{13}^{(1)} + K_{12}^{(1)}K_{14}^{(1)}\right) = 0 \tag{12a}$$

$$Z_{22} \to H\left(K_{22}^{H} + K_{22}^{(1)}\right) + K_{24}^{(1)}\left(K_{12}^{(1)}K_{14}^{(1)} - K_{11}^{(1)}K_{24}^{(1)}\right) - K_{14}^{(1)}\left(K_{22}^{(1)}K_{14}^{(1)} - K_{12}^{(1)}K_{24}^{(1)}\right) = 0 \tag{12b}$$

$$Z_{12} \to H\left(K_{12}^{H} - K_{12}^{(1)}\right) - K_{14}^{(1)}\left(K_{12}^{(1)}K_{14}^{(1)} - K_{11}^{(1)}K_{24}^{(1)}\right) - K_{13}^{(1)}\left(K_{22}^{(1)}K_{14}^{(1)} - K_{12}^{(1)}K_{24}^{(1)}\right) = 0. \tag{12c}$$

The subsidiary condition $H \neq 0$ at the double roots is necessary because $\mathbf{K}_{11}$ in Eq. (11) cannot be singular when $\mathbf{Z}$ is null—if it were, the above expressions would cease to be valid.

Now, although all three of the conditions for double roots given in (12a–c) are known in closed-form, the resulting expressions are far too complicated to be reduced into simple, manageable terms amenable to direct interpretation, not to mention that the eigenvalues remain hidden as arguments in trigonometric functions, i.e. the associated eigenvalue problems are transcendental. This remains true even when these expressions are evaluated and "simplified" with the help of the symbolic tool in Matlab. However, the following is already clear: each of the three conditions in (12a–c) is associated with spectral curves $Z_{11}(\omega, k) = 0, Z_{22}(\omega, k) = 0, Z_{12}(\omega, k) = 0$. Thus, we can envision that the spectral curves of the first set may intersect with those of the second $Z_{11} \cap Z_{22}$ at infinitely many discrete locations, and the same can be said of the first and third set i.e. $Z_{11} \cap Z_{12}$ (or if preferred $Z_{12} \cap Z_{22}$). The likelihood that any of these two sets of intersection points will coincide exactly in space (or even just at a finite subset of points) appears to be low indeed, for these points do not exhibit regular patterns as they do for homogeneous plates with various simple boundary conditions, but are incommensurate instead. We thus posit the strong conjecture that the spectral curves for a system composed of a single layer over an elastic half-space will never exhibit intersections, and by extension, we theorize that this will be true also of any non-symmetric system of layers.

Observe that the case of multiple layers is included in the above. It suffices to replace the elements K_{ij}^{H} by the impedance of the layered system existing below the first layer, as was done for the Love wave case. Of course, in that case the impedances would be known only numerically and not in closed form.

The case of a symmetric structure is an exception to the above because it can then be broken down into two half-spaces (or plates) with complementary mixed boundary conditions at the upper surface. These two cases being independent, their spectral curves can (and indeed do) cross. Observe that in that case we cannot invoke Rayleigh's principle used earlier for the SH wave case. This is because the changeover from a CF (clamped–free) boundary condition to an FC condition represents the simultaneous addition of a constraint combined with a relaxation, so the *interlacing theorem* does not apply.

10. Conclusions

In this article we analyzed in detail the problem of the intersections of the spectral lines of the following systems: a doubly clamped, homogeneous plate; a plate with mixed boundary conditions; a homogeneous free plate; a homogeneous stratum; a non-homogeneous free plate (both symmetric and non-symmetric with respect to its mid-horizon); a layered half-space; and finally a symmetrically layered full space.

The free plate can be seen as the extreme form of a layer over an elastic half-space where the stiffness of the latter is zero, while the stratum would correspond to an infinite rigidity. In the case of the single-layer, homogeneous systems (plate or stratum) we were able to obtain the intersections in closed form. These analyses led us to the following conclusions:

- We have rigorously demonstrated that the spectra for generalized Love (SH) waves in either layered plates or layered half-spaces will never intersect, not even when their material and geometric configurations should be symmetric with respect to some horizontal plane.
- We posit the strong conjecture that non-symmetric half-spaces, strata or plates subjected to SVP waves will *never* exhibit double roots anywhere. However, there will be an abundance of points that masquerade as intersections which ultimately prove to be osculations, i.e. near misses.
- A notable exception to the above rule is the homogeneous stratum when Poisson's ratio is either exactly equal to $1/4$, or it is less than $1/3$ and it satisfies either one of the rational expressions (8a) or (9a), in which case the stratum exhibits infinitely many double roots. However, even in this special case the double roots *coincide* with a subset of the double roots for a doubly-clamped plate, which is fully symmetric, so ultimately the homogeneous stratum is not quite an exception to the rule. No double roots exist in the stratum if Poisson's ratio is either an irrational number, or a rational number that does not satisfy either of these two equations. In those cases, the spectral lines exhibit strong osculations at the spectral points that would have been double roots for any of the nearby Poisson's ratio that do satisfy these two equations.
- A true exception to the previous rule on symmetry may occur when in a layered medium of finite depth, a dilatational and a shear cutoff frequency at zero wavenumber should coincide. In that case, a double root will exist at the double cutoff frequency, whether or not the system is symmetric. Double cutoff frequencies are rare when the system is not homogeneous, because then the sets of S and P cutoff frequencies are both unevenly spaced and incommensurate, so even if one frequency should fortuitously coincide, it is unlikely that further coincident cutoff frequencies will be found. (Note: half-spaces and full spaces do not possess cutoff frequencies.)
- Symmetric structures involving SVP waves will always have double roots, and except for the homogeneous free plate and the homogeneous doubly clamped plate which exhibit myriad intersections, they may also have osculation points. Still, when the material properties of such a symmetric system are slightly altered in a non-symmetrical fashion, the intersections become strong osculations.

Appendix A. Impedance matrices

h = layer thickness; ω = frequency
k = horizontal wavenumber μ = shear modulus
α = P-wave velocity β = S-wave velocity

$$p = \sqrt{1-\left(\frac{\omega}{k\alpha}\right)^2} \qquad s = \sqrt{1-\left(\frac{\omega}{k\beta}\right)^2}$$

$$C_p = \cosh kph\,, \qquad C_s = \cosh ksh$$
$$S_p = \sinh kph\,, \qquad S_s = \sinh ksh$$

SH Waves: (see equation 1)

Layer:

$$k_{11} = ks\mu\frac{C_s}{S_s}, \qquad k_{12} = -ks\mu\frac{1}{S_s} \qquad \mathbf{K}_{SH} = \begin{Bmatrix} k_{11} & k_{12} \\ k_{12} & k_{11} \end{Bmatrix}$$

Half-space impedance: $k_H = ks\mu$

SV-P waves: (see equation 2)

$$K_{11} = 2k\mu\left(\frac{1-s^2}{2D}\right)\frac{C_pS_s - ps\,C_sS_p}{s}$$
$$K_{12} = 2k\mu\left\{\frac{1-s^2}{2D}\left(1-C_pC_s + ps\,S_pS_s\right)+\frac{1+s^2}{2}\right\}$$
$$K_{22} = 2k\mu\left(\frac{1-s^2}{2D}\right)\frac{C_sS_p - ps\,C_pS_s}{p}$$
$$K_{13} = 2k\mu\left(\frac{1-s^2}{2D}\right)\frac{ps\,S_p - S_s}{s}$$
$$K_{14} = 2k\mu\left(\frac{1-s^2}{2D}\right)\left(C_p - C_s\right)$$
$$K_{24} = 2k\mu\left(\frac{1-s^2}{2D}\right)\frac{ps\,S_s - S_p}{p}$$

$$\mathbf{K}_{SVP} = \left\{\begin{array}{cc:cc} K_{11} & K_{12} & K_{13} & K_{14} \\ K_{12} & K_{22} & -K_{14} & K_{24} \\ \hdashline K_{13} & -K_{14} & K_{11} & -K_{12} \\ K_{14} & K_{24} & -K_{12} & K_{22} \end{array}\right\}$$

$$D = \left(\frac{1}{ps}+ps\right)S_pS_s + 2\left(1-C_pC_s\right)$$
$$= \left(\frac{1}{ps}+ps\right)S_pS_s - \left(C_p-1\right)\left(C_s+1\right) - \left(C_p+1\right)\left(C_s-1\right)$$

Half-space impedance:

$$\mathbf{K}_H = 2k\mu\left[\frac{1-s^2}{2(1-ps)}\begin{Bmatrix} p & -1 \\ -1 & s \end{Bmatrix} + \begin{Bmatrix} 0 & 1 \\ 1 & 0 \end{Bmatrix}\right]$$

Appendix B. Double roots of fully clamped plate

Consider a homogeneous layer of total thickness $2h$ which is clamped (i.e. fixed) at both its upper and lower bounding surfaces. Being symmetric, we can choose to solve instead the problem of a layer of thickness h which is clamped at the base and has mixed boundary conditions at the upper surface, first with horizontal rollers and then with vertical rollers. With reference to Eq. (2) and the impedance (stiffness) matrices for SV-P waves given in Appendix A, it is clear that the two complementary problems are each characterized by a single degree of freedom at the upper surface (i.e. either u_{x1} with

$u_{z1} = u_{x2} = u_{z2} = 0$; or u_{z1} with $u_{x1} = u_{x2} = u_{z2} = 0$) for which the characteristic equations are simply

$$K_{11} = 0 \to k\frac{1-s^2}{D}\frac{C_P S_s - ps\, C_s S_p}{s} = 0 \quad \text{Symmetric modes} \tag{B.1}$$

$$K_{22} = 0 \to k\frac{1-s^2}{D}\frac{C_s S_p - ps\, C_p S_s}{p} = 0 \quad \text{Anti-symmetric modes} \tag{B.2}$$

where we have omitted the constant factor $\mu \neq 0$. These two equations define the various branches of the $\omega - k$ spectrum for the symmetric and anti-symmetric modes, and these will intersect at a rich set of discrete points, as will be seen. Alternatively, these intersection points can be understood as the double roots of the fully clamped plate for which Eqs. (B.1), (B.2) must be satisfied simultaneously.

We start by considering the trivial solution $k = 0$ for which we know ahead of time that the solution to (B.1), (B.2) consists of the set of horizontal and vertical cutoff frequencies of the layer with the two alternative sets of mixed boundary conditions, and these are known in closed form, namely

$$\omega_H = \pi\left(n - \tfrac{1}{2}\right)\frac{\beta}{h}, \qquad \omega_V = \pi m\frac{\alpha}{h}, \quad m, n = 1, 2, \ldots \text{ horizontal rollers} \tag{B.3a}$$

$$\omega_H = \pi n\frac{\beta}{h}, \qquad \omega_V = \pi\left(m - \tfrac{1}{2}\right)\frac{\alpha}{h}, \quad m, n = 1, 2, \ldots \text{ vertical rollers.} \tag{B.3b}$$

These cutoff frequencies will produce double roots only when the wave velocities are in some appropriate fractional ratio. It can also be shown (Kausel, 2012) [28] that the coinciding branches at these double points do *not* have zero group velocity as they would normally have at the simple (non-repeated) cutoff frequencies, but have distinct non-zero slopes, one positive and the other negative. For now, we choose to ignore any such double roots.

We also reject the possibility that either of the numbers p or s could vanish because that would represent plane P or S waves propagating horizontally, both of which are forbidden on account of the boundary conditions at the base. Finally, we also disregard the possibility that $1 - s^2 = 0$ i.e. $s = \pm 1$ because that would require $\omega/k = 0$, i.e. zero frequency at non-zero wavenumber (this would require strain energy without kinetic energy). Therefore, the only admissible simultaneous solutions are defined by

$$C_P S_s - ps\, C_s S_p = 0 \tag{B.4a}$$

$$C_s S_p - ps\, C_p S_s = 0. \tag{B.4b}$$

Substituting the second equation into the first, we obtain

$$\left[1 - (ps)^2\right] C_p S_s = 0 \tag{B.5}$$

which demands that either $1 - (ps)^2 = 0$ or $C_p S_s = 0$, the latter of which implies in turn $C_s S_p = 0$.

(a) Case 1: First factor in (B.5) vanishes

The first possibility is that $(ps)^2 = 1$, and from the definition of the numbers p, s we infer

$$(ps)^2 = \left(1 - \left(\frac{\omega}{k\alpha}\right)^2\right)\left(1 - \left(\frac{\omega}{k\beta}\right)^2\right) = 1$$

which implies

$$\left(\frac{\omega}{k\alpha}\right)^2\left(\frac{\omega}{k\beta}\right)^2 - \left(\frac{\omega}{k\alpha}\right)^2 - \left(\frac{\omega}{k\beta}\right)^2 = 0 \quad \text{or} \quad \left(\frac{\omega}{k}\right)^2\left[\left(\frac{\omega}{k}\right)^2 - \left(\alpha^2 + \beta^2\right)\right] = 0$$

so

$$\frac{\omega}{k} = \sqrt{\alpha^2 + \beta^2}. \tag{B.6}$$

It follows that

$$p = \sqrt{1 - \frac{\alpha^2+\beta^2}{\alpha^2}} = \sqrt{\frac{-\beta^2}{\alpha^2}} = \mathrm{i}\,\tilde{p} = \mathrm{i}\,\frac{\beta}{\alpha}, \quad \tilde{p} = \frac{\beta}{\alpha} \tag{B.7a}$$

$$s = \sqrt{1 - \frac{\alpha^2+\beta^2}{\beta^2}} = \sqrt{\frac{-\alpha^2}{\beta^2}} = \mathrm{i}\,\tilde{s} = \mathrm{i}\,\frac{\alpha}{\beta}, \quad \tilde{s} = \frac{\alpha}{\beta} \tag{B.7b}$$

from where we conclude that $ps = -1$ (we exclude the other possibility for the sign of $\tilde{p}$, $\tilde{s}$ because from analytic continuation, the imaginary part of p, s must be positive when $k > 0$, $\omega > 0$). Substituting this result into Eq. (B.4a), we infer that $C_p S_s + C_s S_p = 0$, or in full

$$\sinh ksh\, \cosh kph + \cosh ksh\, \sinh kph = 0$$

i.e.

$$\sinh\left[kh\,(s+p)\right] = \mathrm{i}\,\sin\left[kh\left(\tilde{s}+\tilde{p}\right)\right] = \mathrm{i}\,\sin\left[kh\left(\frac{\alpha}{\beta}+\frac{\beta}{\alpha}\right)\right] = 0$$

which admits the solution $kh\left(\frac{\alpha}{\beta}+\frac{\beta}{\alpha}\right) = j\pi$. Together with (B.7), this implies that

$$kh = j\pi\,\frac{\alpha\beta}{\alpha^2+\beta^2}, \qquad \frac{\omega h}{\beta} = j\pi\,\frac{\alpha}{\sqrt{\alpha^2+\beta^2}}, \quad j = 1, 2, \ldots \tag{B.8}$$

where we have ruled out $j = 0$ to avoid the origin of the spectrum. This set of double roots has the constant phase velocity

$$c = \frac{\omega}{k} = \sqrt{\alpha^2+\beta^2} = \beta\sqrt{\frac{3-4\nu}{1-2\nu}}. \tag{B.9}$$

(b) Second factor in (B.5) vanishes

Now $C_pS_s = 0$ and therefore also $C_sS_p = 0$. One way that this can be satisfied is if $C_p = 0$ and either $C_s = 0$ or $S_p = 0$, but the latter is impossible because a sine and cosine of one and the same argument cannot be simultaneously zero. Alternatively, we could apply the same reasoning to S_P, S_s. Hence, we conclude that either of the following conditions must hold:

$$C_p = 0 \quad \text{and} \quad C_s = 0 \tag{B.10}$$

$$S_p = 0 \quad \text{and} \quad S_s = 0. \tag{B.11}$$

Case 2: Condition (B.10)

Since $C_p = \cosh(kph)$ and $C_s = \cosh(ksh)$, the only way that either $C_p = 0$ or $C_s = 0$ is for p and s to be purely imaginary, say $p = \mathrm{i}\,\tilde{p}$, $s = \mathrm{i}\,\tilde{s}$, where $\tilde{p}$, $\tilde{s}$ are real, positive numbers. It follows that

$$C_p = \cosh(kph) = \cos\left(k\tilde{p}h\right) = 0, \qquad k\tilde{p}h = \left(m-\tfrac{1}{2}\right)\pi \Rightarrow p = \mathrm{i}\,\left(m-\tfrac{1}{2}\right)\frac{\pi}{kh}$$

$$C_S = \cosh(ksh) = \cos\left(k\tilde{s}h\right) = 0 \qquad k\tilde{s}h = \left(n-\tfrac{1}{2}\right)\pi \Rightarrow s = \mathrm{i}\,\left(n-\tfrac{1}{2}\right)\frac{\pi}{kh}$$

$$S_p = \sinh kph = \mathrm{i}\,\sin\left(m-\tfrac{1}{2}\right)\pi = \mathrm{i}\,(-1)^{m-1}$$

$$S_s = \sinh ksh = \mathrm{i}\,\sin\left(n-\tfrac{1}{2}\right)\pi = \mathrm{i}\,(-1)^{n-1}$$

where $m, n = 1, 2, \ldots$ need not be the same *positive* integers. Now, p, s must at the same time satisfy the conditions

$$p^2 = 1-\left(\frac{\omega}{k\alpha}\right)^2 = -\left(m-\tfrac{1}{2}\right)^2\left(\frac{\pi}{kh}\right)^2 \quad \text{and} \quad s^2 = 1-\left(\frac{\omega}{k\beta}\right)^2 = -\left(n-\tfrac{1}{2}\right)^2\left(\frac{\pi}{kh}\right)^2$$

that is

$$(kh)^2 = \left(\frac{\omega h}{\alpha}\right)^2 - \pi^2\left(m-\tfrac{1}{2}\right)^2 \quad \text{and} \quad (kh)^2 = \left(\frac{\omega h}{\beta}\right)^2 - \pi^2\left(n-\tfrac{1}{2}\right)^2. \tag{B.12}$$

For the preceding two equations to be simultaneously true, we would have to have

$$\left(\frac{\omega h}{\beta}\right)^2 = \pi^2\,\frac{\left(n-\frac{1}{2}\right)^2 - \left(m-\frac{1}{2}\right)^2}{1-(\beta/\alpha)^2} \tag{B.13}$$

and since $1-\left(\frac{\beta}{\alpha}\right)^2 = 1/\left[2(1-\nu)\right]$, then from (B.12) and (B.13) we obtain the solution

$$\left(\frac{\omega h}{\beta}\right)^2 = 2\,(1-\nu)\,\pi^2\left[\left(n-\tfrac{1}{2}\right)^2 - \left(m-\tfrac{1}{2}\right)^2\right], \quad m, n = 1, 2, 3, \ldots \tag{B.14a}$$

$$\begin{aligned}(kh)^2 &= \pi^2\left[(1-2\nu)\left(n-\tfrac{1}{2}\right)^2 - 2\,(1-\nu)\left(m-\tfrac{1}{2}\right)^2\right] \\ &= 2\,(1-\nu)\,\pi^2\left[\left(\frac{\beta}{\alpha}\right)^2\left(n-\tfrac{1}{2}\right)^2 - \left(m-\tfrac{1}{2}\right)^2\right]\end{aligned} \qquad \frac{n-\frac{1}{2}}{m-\frac{1}{2}} > \frac{\alpha}{\beta}. \tag{B.14b}$$

In Eqs. (B.14a,b), we should rule out the possibility that $ps = -1$, because if allowed, that would repeat some of the roots already found. With reference to Eq. (B.9), this would require

$$\left(\frac{\omega}{k\beta}\right)^2 = \frac{\left(n-\frac{1}{2}\right)^2 - \left(m-\frac{1}{2}\right)^2}{\left(\frac{\beta}{\alpha}\right)^2\left(n-\frac{1}{2}\right)^2 - \left(m-\frac{1}{2}\right)^2} \neq \frac{\alpha^2+\beta^2}{\beta^2}$$

which after brief algebra leads us to

$$\frac{2n-1}{2m-1} \neq \left(\frac{\alpha}{\beta}\right)^2 = \frac{2-2\nu}{1-2\nu}$$

or

$$\nu \neq \frac{2n-4m+1}{4\,(n-m)}. \tag{B.15}$$

Clearly, if the actual Poisson's ratio were to agree with the right hand side of (B.15), that would guarantee that $ps = -1$, in which case the solution would be invalid.

Case 3: Condition (B.11)

Once more we conclude that p, s must be purely imaginary numbers, in which case

$$S_p = \sinh(kph) = \mathrm{i}\,\sin\left(k\tilde{p}h\right) = 0 \qquad k\tilde{p}h = m\pi \Rightarrow p = \mathrm{i}\,m\frac{\pi}{kh}$$

$$S_s = \sinh(ksh) = \mathrm{i}\,\sin\left(k\tilde{s}h\right) = 0 \qquad k\tilde{s}h = n\pi \Rightarrow s = \mathrm{i}\,n\frac{\pi}{kh}$$

$$C_p = \cosh(kph) = \cos m\pi = (-1)^m$$

$$C_s = \cosh(ksh) = \cos n\pi = (-1)^n$$

$$ps = -mn\left(\frac{\pi}{kh}\right)^2$$

where in principle m, n could be any non-negative integers. However, we already established that $p \neq 0$ and $s \neq 0$, which means the values $m = 0$, $n = 0$ must be rejected, in which case these parameters must be positive integers. It follows that

$$p^2 = 1 - \left(\frac{\omega}{k\alpha}\right)^2 = -m^2\left(\frac{\pi}{kh}\right)^2 \quad \text{and} \quad s^2 = 1 - \left(\frac{\omega}{k\beta}\right)^2 = -n^2\left(\frac{\pi}{kh}\right)^2$$

so

$$(kh)^2 = \left(\frac{\omega h}{\alpha}\right)^2 - \pi^2 m^2 \quad \text{and} \quad (kh)^2 = \left(\frac{\omega h}{\beta}\right)^2 - \pi^2 n^2$$

implying

$$\left(\frac{\omega h}{\beta}\right)^2 = \pi^2\frac{n^2-m^2}{1-(\beta/\alpha)^2} \tag{B.16}$$

so with $1 - \left(\frac{\beta}{\alpha}\right)^2 = 1/[2(1-\nu)]$ we obtain the solution

$$\left(\frac{\omega h}{\beta}\right)^2 = 2\,(1-\nu)\,\pi^2\left(n^2-m^2\right) \geq 0 \quad m, n = 1, 2, 3\ldots \tag{B.17a}$$

$$\begin{aligned}(kh)^2 &= \pi^2\left[(1-2\nu)\,n^2 - 2\,(1-\nu)\,m^2\right] \\ &= 2\,(1-\nu)\,\pi^2\left[\left(\frac{\beta}{\alpha}\right)^2 n^2 - m^2\right] \geq 0 \quad n > \frac{\alpha}{\beta}m.\end{aligned} \tag{B.17b}$$

Again, we must rule out the possibility that (B.17a,b) may include roots for which $ps = -1$. Following the same approach as in the previous section, it can be shown that this would require

$$\frac{n}{m} \neq \left(\frac{\alpha}{\beta}\right)^2 = \frac{2-2\nu}{1-2\nu}$$

or

$$\nu \neq \frac{n-2m}{2\,(n-m)}. \tag{B.18}$$

Once more, we would need to reject this solution if the actual Poisson's were to agree with (B.18).

In conclusion, Eqs. (B.8), (B.14) and (B.17) define the complete set of intersections (i.e. double roots) of the spectral lines of the double-thickness, doubly clamped plate. These are summarized again in Eqs. (5a), (5b), (5c) in the main text.

Appendix C. Double roots of homogeneous stratum

Consider a homogeneous stratum of thickness h, i.e. a layer with a free surface which is underlain by a rigid (clamped) boundary at the bottom. This system is characterized by two degrees of freedom at the surface, and its characteristic eigenvalue problem is (see also Eq. (2) and Appendix A)

$$\begin{vmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{vmatrix} = 0.$$

Any double roots of the stratum will cause the above system to have a loss of rank of two, which requires the simultaneous satisfaction of the three conditions $K_{11} = K_{22} = K_{12} = 0$. Since the conditions $K_{11} = K_{22} = 0$ are already satisfied by the doubly-clamped plate given in Appendix B, we see that these double roots are constituted by a subset of the intersections of that plate, namely the subset which satisfies the additional condition $K_{12} = 0$:

$$K_{12} = 0 \to \frac{1-s^2}{D}\left(1 - C_pC_s + psS_pS_s\right) + \left(1+s^2\right) \stackrel{?}{=} 0. \tag{C.19}$$

However now, because D does not divide both terms, we must also require $D \neq 0$, which will rule out some of the previous solutions. This is satisfied if

$$D = \left(\frac{1}{ps} + ps\right) S_pS_s + 2\left(1 - C_pC_s\right) \neq 0. \tag{C.20}$$

With the three solutions previously obtained for the fully clamped plate, this results in the following:

(a) Case 1

$$ps = -1$$

$$D = 2\left(1 - S_pS_s - C_pC_s\right) = 2\left[1 - \cosh\left(p+s\right)kh\right] = 2\left[1 - \cos\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)kh\right]$$

$$= 2\left[1 - \cos j\pi\right] = \begin{cases} 4 & j \text{ is odd} \\ 0 & j \text{ is even.} \end{cases}$$

We see that we must rule out even indices j, which is most readily accomplished by changing j into $2j-1$. Also, since $1 - C_pC_s + psS_pS_s = 1 - \cos\left(2j-1\right)\pi = 2$, then

$$K_{12} = 0 \to \frac{1-s^2}{4}2 + \left(1+s^2\right) = \frac{1}{2}\left(3 - s^2\right) = 0$$

which can *only* be satisfied if $s = \mathrm{i}\sqrt{3}$ =. Also, from Eq. (B.7b), we also know that $s = \mathrm{i}\sqrt{\frac{\alpha}{\beta}}$, in which case the coupling term K_{12} can only vanish when $\alpha = \beta\sqrt{3}$, which is when Poisson's ratio equals $\nu = 0.25$. In that particular case, and from (B.8) with the change $j \to 2j-1$, we obtain the double roots

$$kh = \left(2j-1\right)\pi\frac{\sqrt{3}}{4}, \qquad \frac{\omega h}{\beta} = \left(2j-1\right)\pi\frac{\sqrt{3}}{2} \quad j = 1, 2, 3\ldots \qquad \nu = \tfrac{1}{4} \tag{C.21}$$

(b) Case 2

$$C_p = C_s = 0, \qquad S_p = \mathrm{i}\,(-1)^{m-1}, \qquad S_s = \mathrm{i}\,(-1)^{n-1}, \qquad S_pS_s = (-1)^{m+n-1}$$

$$p = \mathrm{i}\,\tilde{p} = \mathrm{i}\left(m - \tfrac{1}{2}\right)\frac{\pi}{kh}, \qquad s = i\,\tilde{s} = \mathrm{i}\left(n - \tfrac{1}{2}\right)\frac{\pi}{kh}$$

$$D = 2 + \left[1/\left(ps\right) + ps\right](-1)^{m+n-1}.$$

Consider next the condition $K_{12} = 0$, which after multiplication of (C.19) by D reduces to the test

$$\left(1-s^2\right)\left(1 + ps\,(-1)^{m+n-1}\right) + \left[\left(\frac{1}{ps} + ps\right)(-1)^{m+n-1} + 2\right]\left(1+s^2\right) \stackrel{?}{=} 0$$

or after some algebra.

$$(1 + 2p^2s^2 + s^2)(-1)^{m+n-1} + (3 + s^2)ps \stackrel{?}{=} 0.$$

Making use of the symbolic tool in Matlab, it can be shown that the above can be reduced to the following conditional tests, both of which must satisfy the condition

$$d = 4n^2 - 8m^2 - 4n + 8m - 1 - 8\nu\left(m + n - 1\right)\left(n - m\right) \neq 0. \tag{C.22}$$

- If $n > m$ is an even–odd pair (i.e. $(-1)^{m+n-1} = 1$), in which case

$$[4\nu\,(m-n) + 2n + 1 - 4m]\,[2\nu\,(m-n) + 1 - 2m] \stackrel{?}{=} 0.$$

This implies either of the identities

$$\nu = \frac{1-2m}{2\,(n-m)} \quad \text{or} \quad \nu = \frac{2n-4m+1}{4\,(n-m)}.$$

For $0 \le \nu \le 1/2$, the first is impossible, while the second is in principle possible. However, in that case the Poisson ratio agrees exactly with the right-hand side of (B.15), which guarantees that $ps = -1$ and $D = 0$. Hence, we must reject this solution.

- If $n > m$ are both even–even or both odd–odd (i.e. $(-1)^{m+n-1} = -1$), then

$$[4\nu\,(n+m-1) - 2n - 4m + 3]\,[2\nu\,(n+m-1) + 1 - 2m] \stackrel{?}{=} 0$$

which implies either of the identities

$$\nu = \frac{2n+4m-3}{4\,(n+m-1)}$$

$$\text{or} \quad \nu = \frac{2m-1}{2\,(n+m-1)} \qquad 0 < \nu < \tfrac{1}{3}. \tag{C.23}$$

The first is impossible because for any legal combination of m, n it would imply a Poisson's ratio greater than 0.5. The second is valid for any pair of even–even or odd–odd integers $[m,\, n > m]$, and in addition it does not violate (B.15). Substituting (C.23) into (C.22), it will also be seen that $d = (2n-1)\,(2n-4m+1) \neq 0$. Hence, no condition is violated, so the solution is legitimate. Now, although it would seem from (C.23) that by proper combination of m, n one could achieve any Poisson's ratio $\nu < \frac{1}{2}$, this is actually not possible because (B.14b) demands that $(2n-1)^2 / (2m-1)^2 > (\alpha/\beta)^2 = (2-2\nu)/(1-2\nu)$. This will show up as an additional restriction on the choice of m, n, namely $n > 2m - \frac{1}{2}$.

Let m, n be any valid pair satisfying the additional restriction that $2m-1,\ 2n-1$ are relatively prime integers, then the value of Poisson's ratio in (C.23) does not change if we generate another pair according to the rule

$$\left[m',\, n'\right] = \left[m + (j-1)\,(2m-1)\,,\, n + (j-1)\,(2n-1)\right], \quad j = 1, 2, 3, \ldots \tag{C.24}$$

which implies

$$\left[2m' - 1,\ 2n' - 1\right] = (2j-1)\,[2m-1,\ 2n-1]$$

then from (B.14a,b), after simplifications

$$kh = \tfrac{1}{2}\pi\,(2j-1)\,\sqrt{(2n-1)\,(2n-4m+1)} \tag{C.25a}$$

$$\frac{\omega h}{\beta} = \pi\,(2j-1)\,\sqrt{(2n-1)\,(n-m)} \tag{C.25b}$$

provided that the sub-radical in (C.25a) is positive, i.e. that $n > 2m - \frac{1}{2}$.

(c) Case 3

$$S_p = S_s = 0, \qquad C_p = (-1)^m\,, \qquad C_s = (-1)^n$$

$$D = 2\left(1 - (-1)^{m+n}\right) = \begin{cases} 0 & m+n \text{ is even} \\ 4 & m+n \text{ is odd.} \end{cases}$$

We see that we must rule out the combinations where m, n are either both even or they are both odd. Also

$$1 - s^2 = 1 + n^2\left(\frac{\pi}{kh}\right)^2, \qquad 1 + s^2 = 1 - n^2\left(\frac{\pi}{kh}\right)^2, \qquad ps = -mn\left(\frac{\pi}{kh}\right)^2$$

$$(kh)^2 = \pi^2\left[(1-2\nu)\,n^2 - 2\,(1-\nu)\,m^2\right]$$

so

$$K_{12} \to \frac{\left[1 + n^2\left(\frac{\pi}{kh}\right)^2\right]\left[1 - (-1)^{m+n}\right]}{2\left[1 - (-1)^{m+n}\right]} + 1 - n^2\left(\frac{\pi}{kh}\right)^2$$

$$= \tfrac{1}{2}\left[1 + n^2\left(\frac{\pi}{kh}\right)^2\right] + 1 - n^2\left(\frac{\pi}{kh}\right)^2$$

$$= \tfrac{3}{2} - \tfrac{1}{2}n^2\left(\frac{\pi}{kh}\right)^2$$

i.e.

$$K_{12} \to 3 - \frac{n^2}{(1-2\nu)\,n^2 - 2\,(1-\nu)\,m^2} \stackrel{?}{=} 0.$$

This will be satisfied if and only if

$$\nu = \frac{n^2 - 3m^2}{3\left(n^2 - m^2\right)} \quad 0 < \nu < \tfrac{1}{3} \tag{C.26}$$

with n even and m odd (or vice-versa), and $n^2 > 3m^2$. This Poisson's ratio does not violate Eq. (B.18), so the solution is valid.

Now, if we require m, n to be a pair of relatively prime numbers and choose an odd factor $\ell = 2j - 1$, then the new pair $m' = \ell m$, $n' = \ell n$ will preserve the odd–even or even–odd relationship of the pair of indices m, n, in which case ℓ^2 factors out of (C.26) and the Poisson's ratio is preserved. Making use of (C.26) in (B.17a,b) we then obtain

$$kh = \pi\,(2j-1)\,n\frac{\sqrt{3}}{3} \qquad \frac{\omega h}{\beta} = \pi n\,(2j-1)\,\frac{2\sqrt{3}}{3}, \quad j = 1, 2, \ldots. \tag{C.27}$$

Summary of results for stratum

From the foregoing we conclude that a homogeneous stratum will exhibit infinite sets of double roots as follows:

(a) When $\nu = \frac{1}{4}$, an infinite number of double roots of the form given by (C.21) exist.
(b) If (m, n) is a pair of even–even or odd–odd integers satisfying the condition $n > m$ and in addition the pair $2m-1,\ 2n-1$ is relatively prime, then there exists an infinite set of double roots of the form (C.24), (C25a,b) whose Poisson's ratio is defined by (C.23) and lies in the open interval $0 < \nu < 1/3$.
(c) If (m, n) is a pair of even–odd or odd–even integers satisfying the condition $n > m\sqrt{3}$ and in addition they are relatively prime, then there exists an infinite set of double roots of the form of (C.27) whose Poisson's ratio is defined by (C.26), and lies on the open interval $0 < \nu < 1/3$.
(d) No double roots exist whenever Poisson's ratio is either a rational number that does not satisfy Eqs. (C.23), (C.26), is an irrational number, or it exceeds $1/3$.

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